



POLITECNICO
MILANO 1863

Nonparametric Statistics Project

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1 Introduction and Scope of the Project

We conducted our analysis on the Annual Developer Survey carried out by Stack Overflow in the years 2022, 2023, and 2024. The dataset contains responses from software developers worldwide regarding their work experience, annual salary, programming languages used, and other professional characteristics. Our focus was specifically on data scientists and machine learning developers with full-time employment in Europe, the US, and Canada, aiming to uncover key trends in their compensation, skills, and work environments.

Given this focus, our research question is:

What are the key factors that influence the salary of a data scientist, how are the most in-demand skills evolving, and what are the future trends in work models?

Our objective is to provide practical, interactive, data-driven insights for companies, in order to better understand compensation benchmarks, work model trends for strategic workforce planning and to identify critical skills to make more informed hiring decisions.

2 Dataset and Pre-Processing

The first step we performed was to retain only the observations relevant to our investigation, removing all the amateur developers or those working in countries outside the scope of our research for each year. Additionally, we narrowed our focus to respondents who have attended university and are employed by a company.

The second step was a preliminary feature selection in which we eliminated all the questions in the survey that were not related to the job-specific characteristics.

Finally we gathered together the observations of the different years, adjusting the salaries with respect to the inflation.

The final dataset is made up of 848 observations and the following 7 variables:

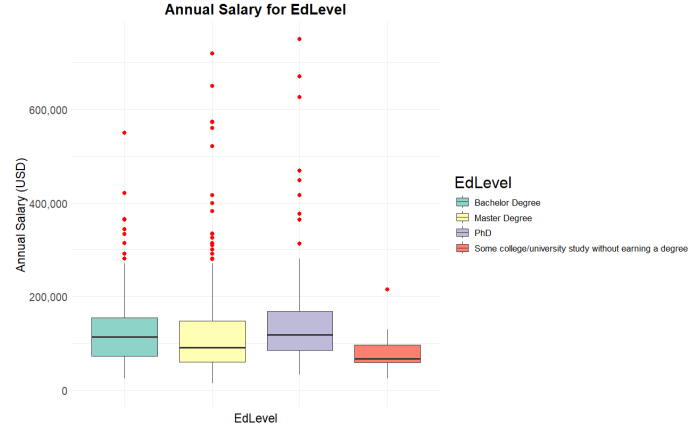
- *RemoteWork*: Categorical variable that indicates if the respondent works remotely, in a hybrid fashion or in person.
- *EdLevel*: Categorical variable that indicates the highest degree obtained by the respondent.
- *OrgSize*: Categorical variable that indicates the range of the number of employees in the respondent's company.
- *Country*: Country where the respondent works.
- *LanguageHaveWorkedWith*: For each respondent this variable contained all the programming languages he/she used. During the preprocessing phase we created 7 binary variables that for each language indicated if the interviewee had worked with it or not. The language considered are *C++*, *Julia*, *MATLAB*, *Python*, *R*, *Scala*, *SQL*.

- *WorkExp*: Years of work experience of the respondent.
- *ConvertedCompYearly*: Annual salary of the respondent converted in US dollars (\$) and adjusted for inflation.

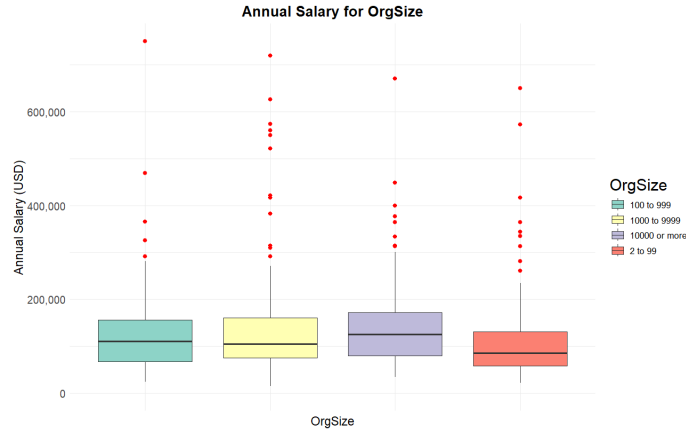
3 Methods

3.1 Visual Exploration

The initial visual exploration of the data was made through boxplots of the salary for the different categories that we show in the following *Figures*.

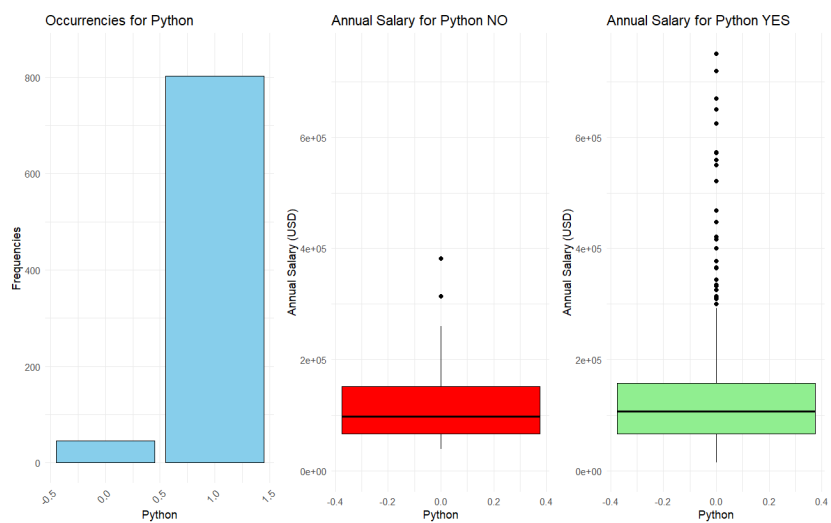


(a) *EdLevel*

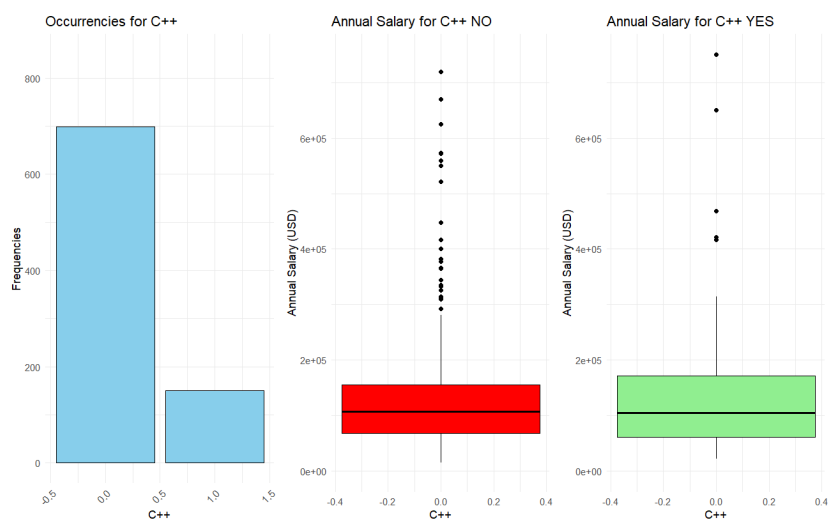


(b) *OrgSize*

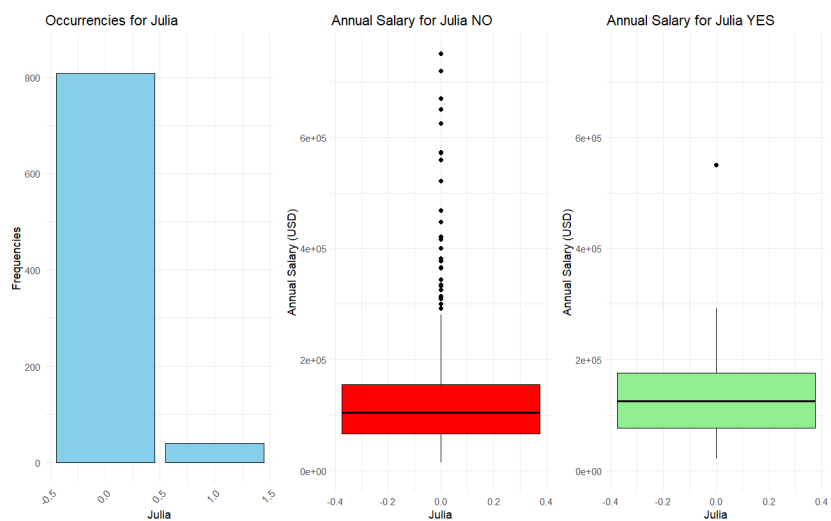
To visualize the knowledge of programming languages we report for each one the frequency of users and the boxplot of the salaries of users and non-users.



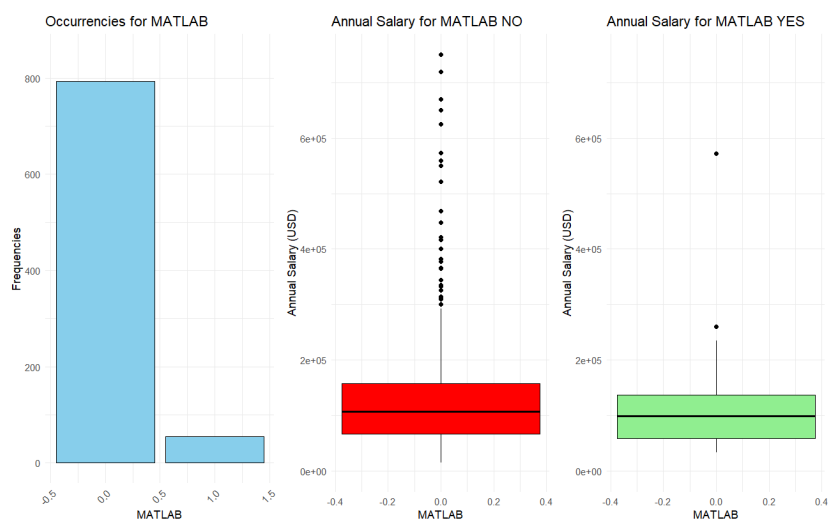
Python



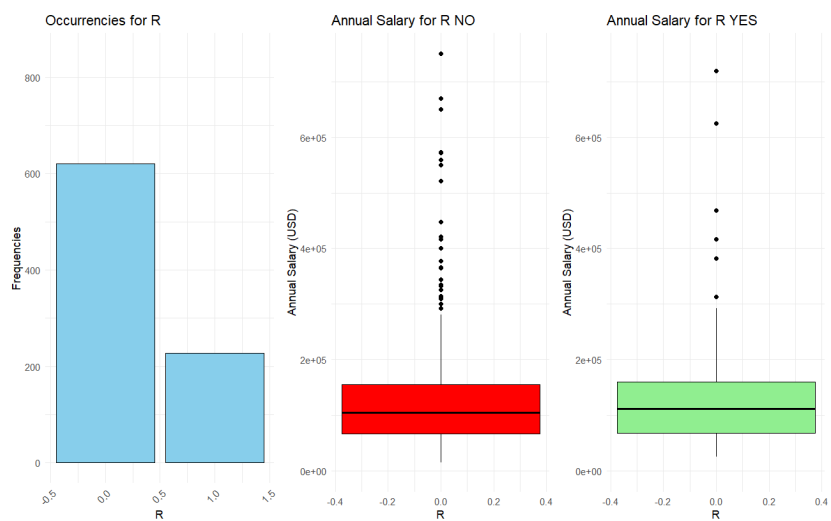
C++



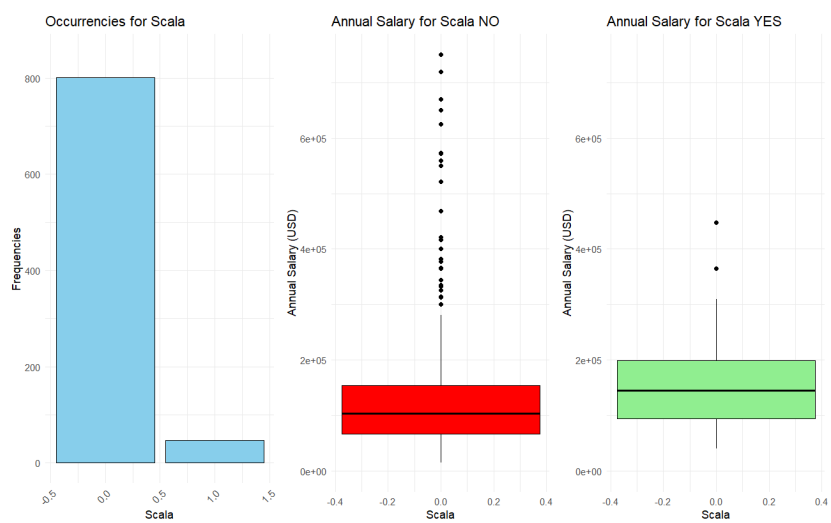
Julia



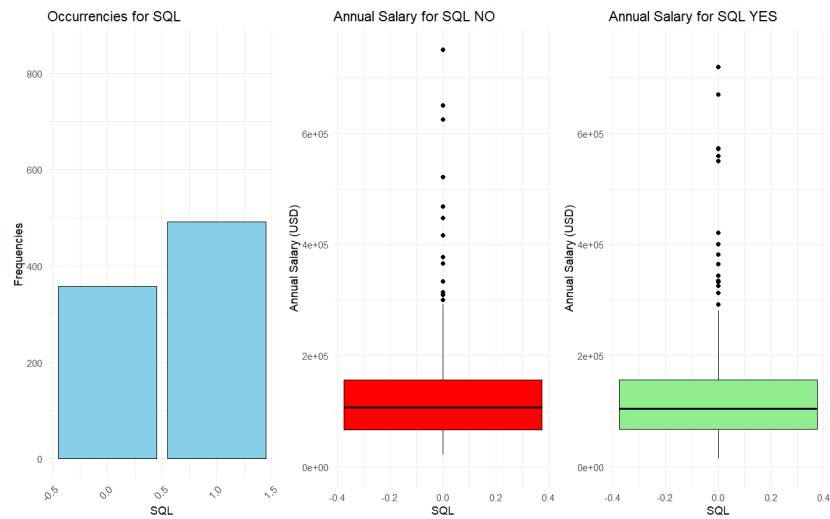
Matlab



R



$Scala$



SQL

We also wanted to visualize the salary together with the years of work experience. Here we show a scatterplot of these two variables along with the regression line.

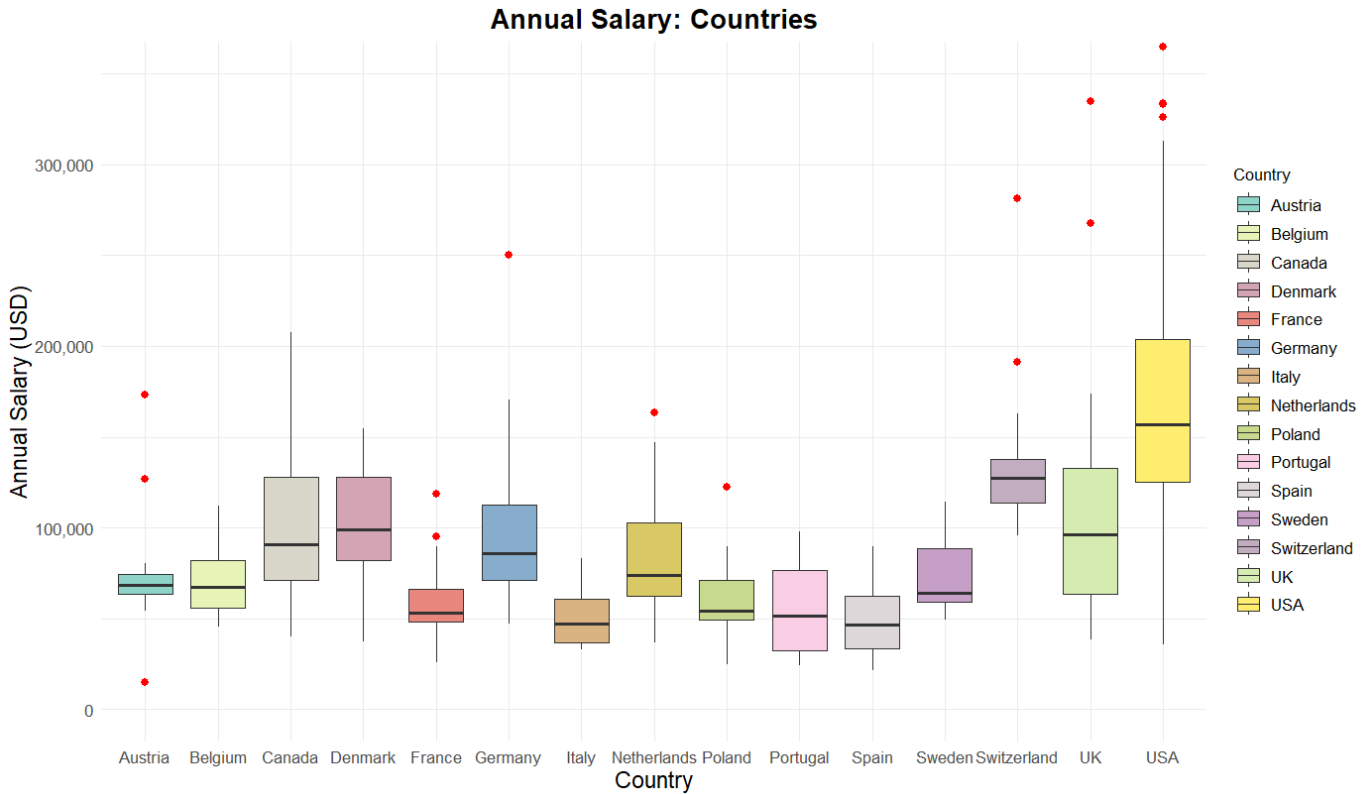


Scatterplot

A simple regression line does not seem to be able to well fit the data, we will try some different models in the following steps.

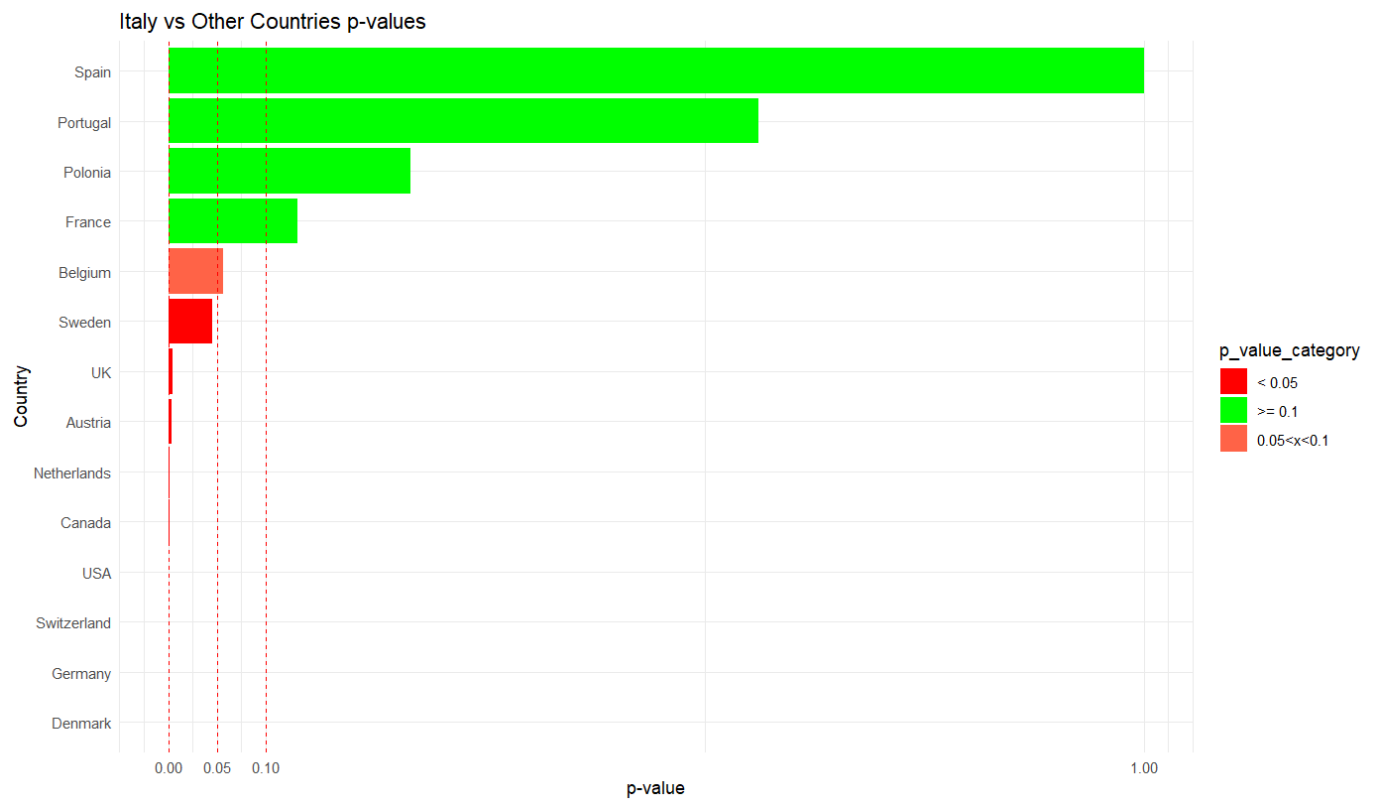
3.2 Focus on Countries

As we did before with other variables, here we report the boxplot of the annual salary grouped by Country.

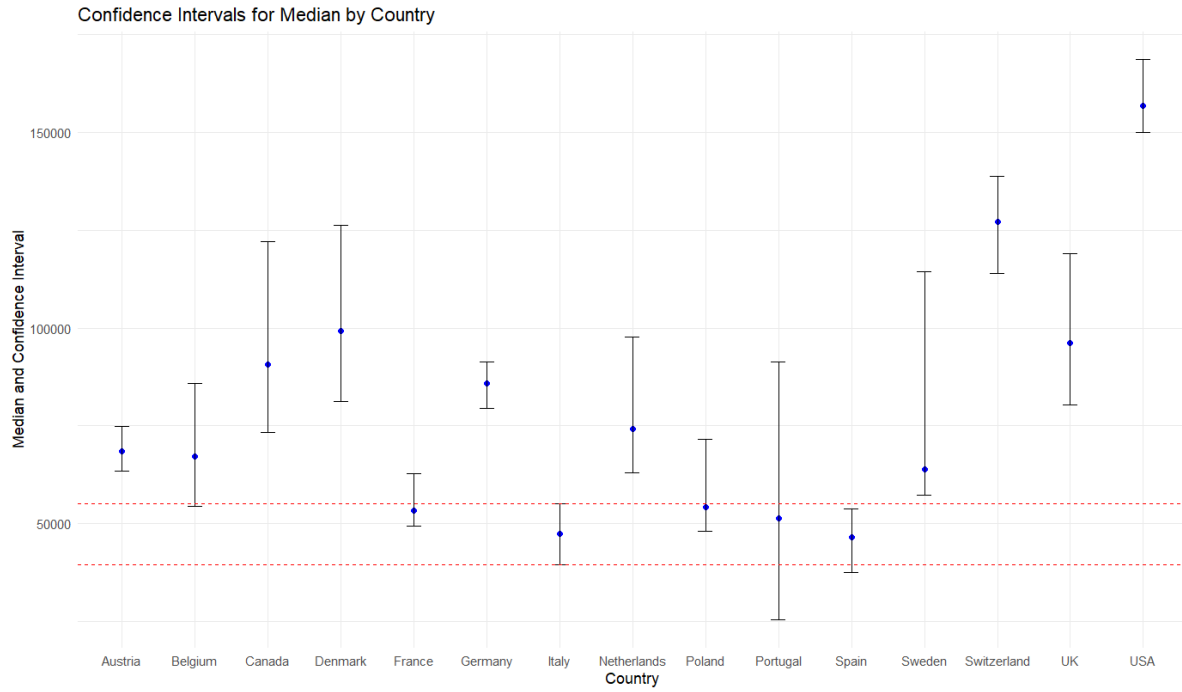


Salaries in the most coveted countries

Among the features, the Country seems to be the one with the most influence on the annual salary. In particular we can see a critical situation regarding Italy, which seems to have one of the lowest distribution of salaries among the considered countries. To check if this difference is actually significant, we performed several **Permutation Tests** to check if the distribution of the salaries in Italy is equal to the distribution of the salaries in the other countries. In the following *Figure* we report the Monte-Carlo estimation of the p-values of these tests.



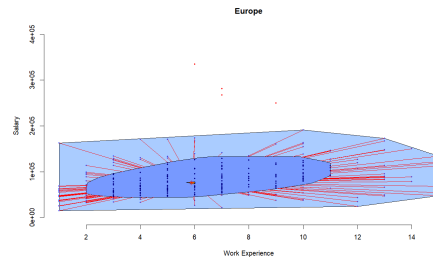
We can see from the *Figure* that there are only a few countries for which we can assume equality of distribution with Italy, while for all the other countries we have to reject the hypothesis. This leads to a bad scenario for Italy nowadays. We then used bootstrap to build confidence intervals for the median of the salaries.



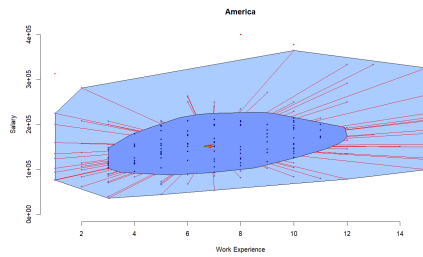
3.3 Further analysis on salary and work experience

In this section we are going to analyze how the work experience impacts on the salary, first analyzing separately the two continents (Europe and America) and then on the entire dataset.

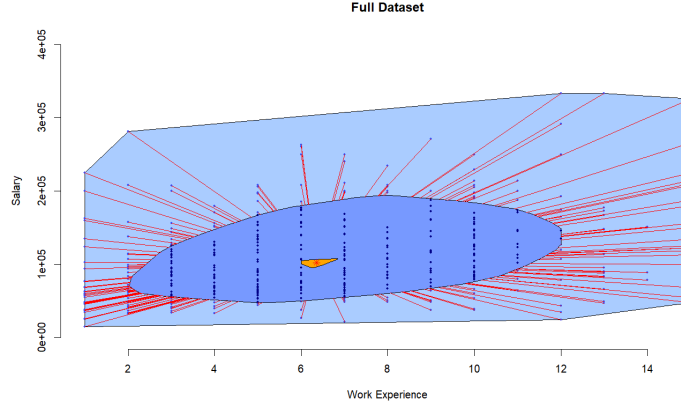
Before starting the analysis we tried to find the possible outliers through bag-plots and decided to remove them.



(c) Europe



(d) America

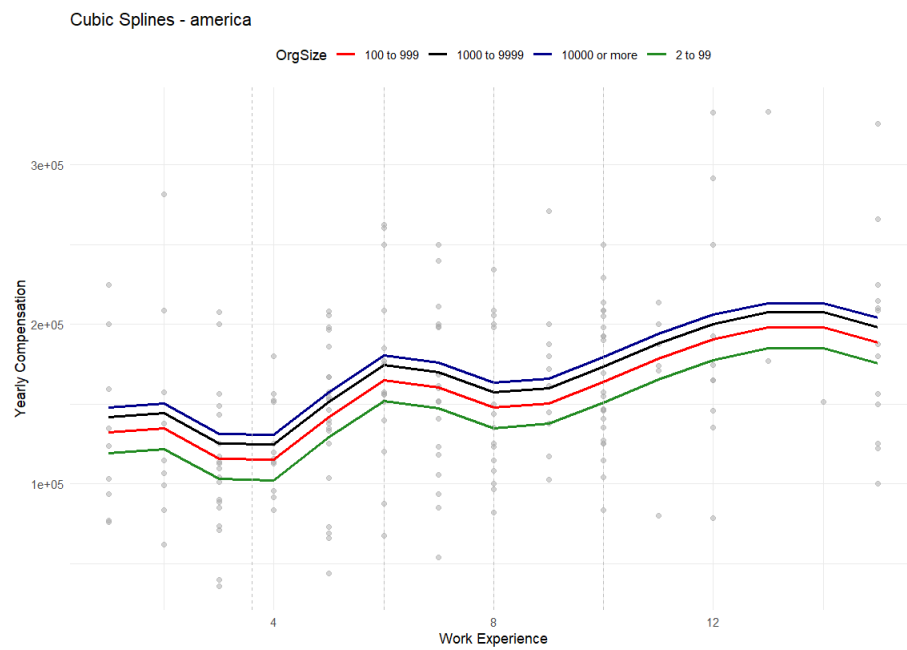
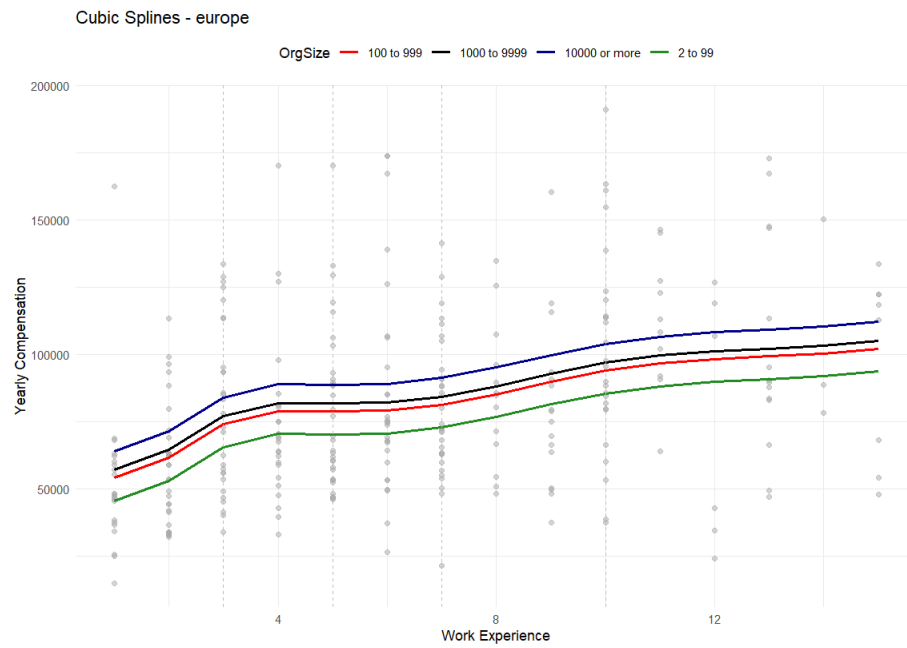


Full Dataset

Based on the plots it seems difficult to fit a good model to predict the annual salary based only on the work experience, so we decided to add to the model also the *Educational Level* of the candidate and the *Organization Size* of the company. Before doing that, we performed a permutational ANOVA to verify if these two variables significantly influenced the salary. We report the estimated p-values in the following *Table*.

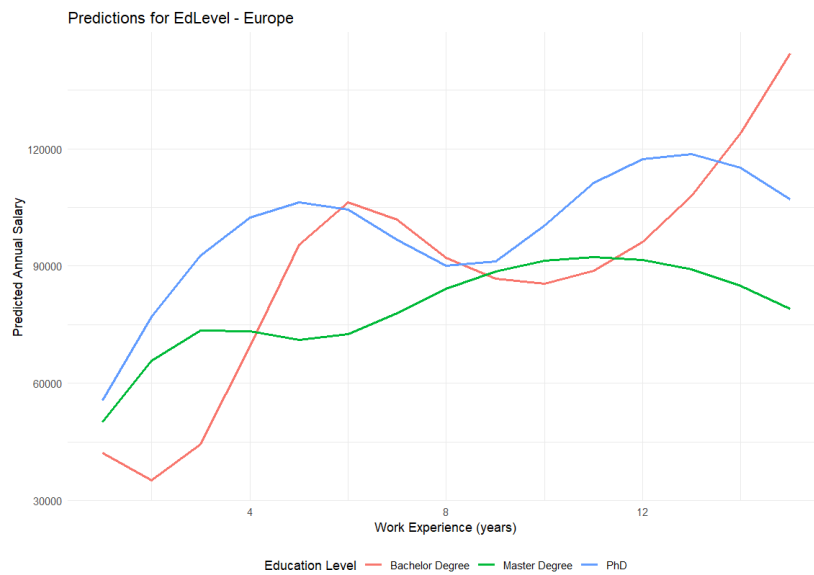
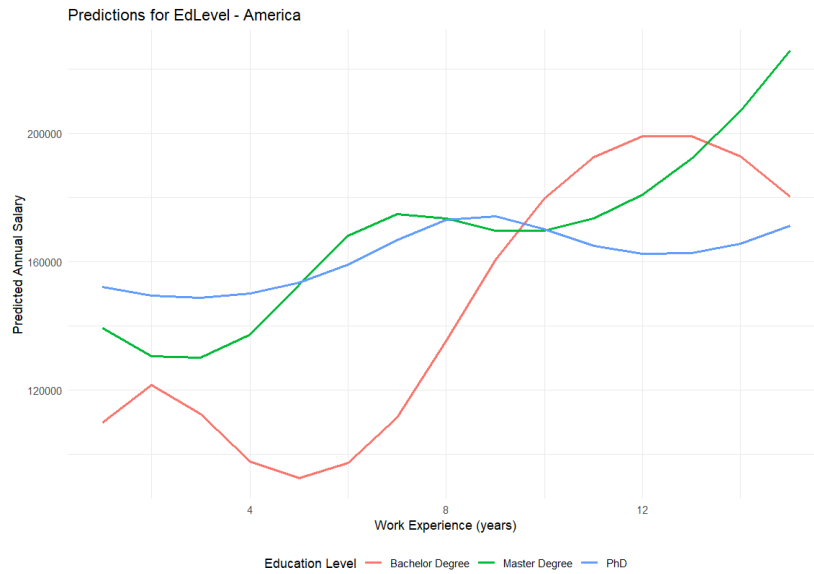
Continent	Variable	P-Value
Europe	EdLevel	0.0000
Europe	OrgSize	0.0002
America	EdLevel	0.0004
America	OrgSize	0.0389
Both	EdLevel	0.0004
Both	OrgSize	0.0001

After evaluating the significance of the influence of the above-mentioned factors, we initially fitted different cubic splines adding the Organization Size as a fixed intercept both for America and Europe.



From the plot we can see that in general a bigger company results in bigger salaries for its employees.

For what regards the education level we were also interested on how it impacted in different times considered, so, instead of adding a fixed intercept, we fitted different splines on the observations that belonged to a specific education level.

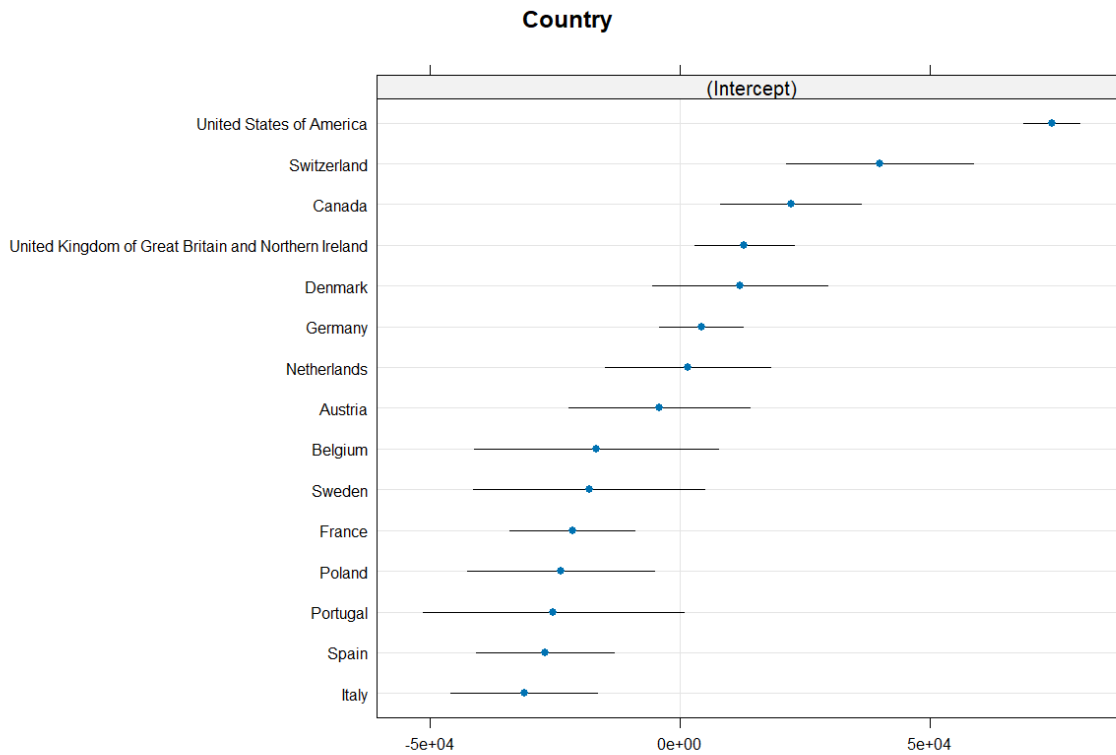


We can see that in the beginning of the career (approximately the first 5 years) an higher degree implies an higher salary. After that the difference seems

to disappear, probably because the skills acquired with the work experience matter more than the study degree.

After these considerations we are ready to present our full model, consisting of the fitting of cubic splines on the work experience, two fixed intercepts representing the educational level and the organization size and finally one random intercept representing the country. We report some indices of goodness of fit and the dotplot of the random intercept.

R^2_{adj}	0.494
PVRE	0.415



Dotplot of random intercept

As we can see the model is far from being perfect, however we believe that it could be useful to give indications about the salaries a company should offer when hiring a data scientist given their individual characteristics and the company size and location.

4 Additional Features

4.1 Coding Association Rules

For this part of the project we investigated the most frequent associations of programming languages known by the respondents. We used the Apriori algorithm that was able to provide combinations of languages that the company could teach to new employees during training or add as a requirement for a job application.

We focused only on associations that presented **lift** > 1 where the **lift** indicates how the presence of a certain language impacts on the probability of observing in the same subject the knowledge of another language.

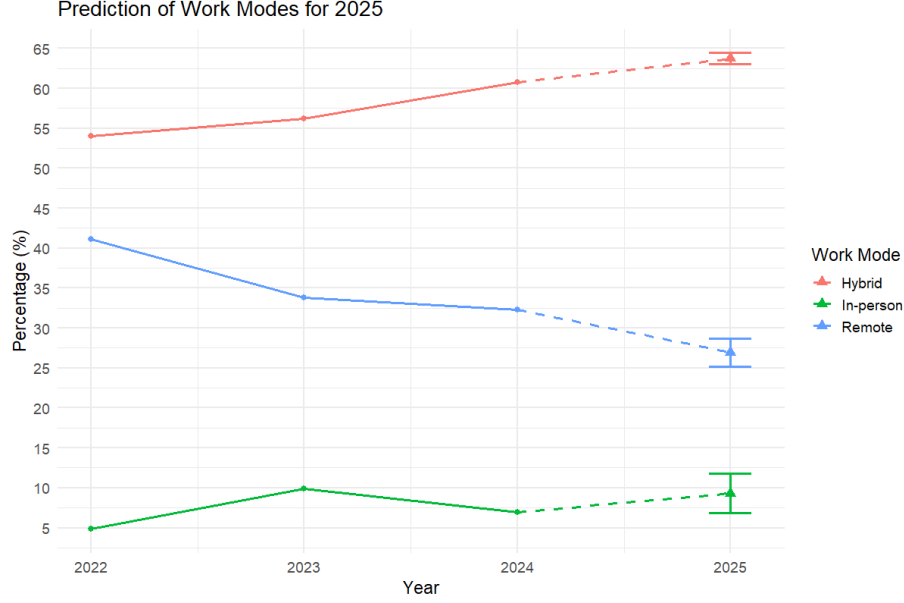
The highest lift was found in the association $\{Python, R\} \implies \{SQL\}$. Here we show the values with **lift** > 1.

	lift
Python,R $\implies$ SQL	1.2976
Python,SQL $\implies$ R	1.1873

As a result, since Python and R are very common languages in data science, we would suggest to include SQL courses or requirements when training or hiring new employees.

4.2 Work Mode Prediction

We also tried to analyze if it could be convenient for a company to encourage the transition to smart working based on last years' trend. To do so we performed a conformal prediction for the year 2025 of the percentages of the different work modes (Remote,In-Person,Hybrid).



$$\alpha = 0.1$$

Level	Prediction2025	LowerBound	UpperBound
Remote	26.937	25.209	28.665
Hybrid	63.747	63.044	64.450
In-Person	9.316	6.885	11.747

In conclusion we see that it seems important to encourage remote working but without completely sacrificing office work.

5 Conclusions

This study analyzed data from the Stack Overflow Developer Survey (2022–2024) to provide insights into salary structures, programming language associations, and work mode trends for data scientists and machine learning developers in Europe, the US, and Canada. The findings highlight the significant impact of country, work experience, education level, and company size on salary levels, offering a data-driven approach for companies to structure competitive compensation packages.

Furthermore, the study identified key programming language associations, emphasizing the importance of SQL training for professionals proficient in Python and R. This insight aids organizations in designing effective employee training programs. Lastly, the prediction of work mode distribution for 2025 suggests

a dominance of hybrid work (63.7%), reinforcing the need for companies to balance remote flexibility with in-office collaboration.

This analysis is not a definitive answer to the complexities of the data science job market but a valuable data-driven approach to existing data. By analyzing salary structures, skill relationships and work models, we provide actionable insights that help companies and recruiters make more effective hiring strategies.